

Statistics (ABC260S) Test 1

Qualification Name	Diploma in Stuff	Qualification Code	D3STUF
Examiner	Mr E Fudd	Date	26 November 2025
Examiner's Department	Mathematics & Physics	Time	09h00 – 11h00
Internal Moderator	Ms M Mouse	Duration	Two Hours
Total Pages	4	Total Marks	37
Instructions			
• You must answer all questions. You may answer the questions in any order. • All answers must be written in the answer booklet provided, not on the question paper. • Show all steps and not only the final answer. • Numerical answers may be expressed as fractions in lowest terms or decimals correct to four significant digits.			

Useful Formulas

$$\Pr(a \leq X \leq b) = \int_a^b f_X(x) dx$$

$$\Pr(a \leq X \leq b) = F_X(b) - F_X(a)$$

$$F_X(x) = \Pr(X \leq x) = \int_{-\infty}^x f_X(t) dt \text{ for } -\infty < x < \infty$$

$$f_X(x) = \frac{dF_X(x)}{dx}$$

Questions

1. [10 marks] Answer the following multiple choice questions (two marks each).

- a. [2 marks] What is the capital city of Morocco?
 - i. Rabat
 - ii. Tunis
 - iii. Casablanca
 - iv. Los Angeles
- b. [2 marks] What is the deepest lake in the world?
 - i. Tanganyika
 - ii. Superior
 - iii. Baikal
 - iv. Argentino
- c. [2 marks] What is $3 + 8$?
 - i. 5
 - ii. 11
 - iii. 24
 - iv. 38
- d. [2 marks] Which of the below countries has the largest population?
 - i. Mexico
 - ii. Nigeria
 - iii. Brazil
 - iv. Indonesia
- e. [2 marks] The most viewed YouTube video of all time is...
 - i. Despacito by Luis Fonsi
 - ii. Gangnam Style by Psy
 - iii. Baby Shark Dance by Pinkfong
 - iv. See You Again by Wiz Khalifa

2. [11 marks] Below are given the final marks of a class of twelve students.

47 40 58 45 70 55 54 47 35 47 31 94

- [3 marks] Find the mean of the final marks.
- [3 marks] Find the median of the final marks.
- [3 marks] Find the standard deviation of the final marks.
- [2 marks] Draw a histogram of the final marks. Use a bin width of 10.

3. [9 marks] Consider the random variable X with probability density function as follows:

$$f_X(x) = \begin{cases} k(x^2 + x) & \text{for } 1 \leq x \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

- [3 marks] Find the value of k .
 - [3 marks] Find $\Pr(X > 2)$.
 - [3 marks] Show the probability just calculated on a graph of $f_X(x)$.
4. [7 marks] Consider the dataset provided in the Excel file `surf4cars_Toyota_Corolla.xlsx`. A description of the columns in the dataset is as follows. Answer the following questions using R code and output only.
- [4 marks] Fit a linear regression model to the data where Asking Price is the response variable and Year Model, Mileage, and Transmission Type are explanatory variables. Display a summary of the model output.
 - [3 marks] Predict the asking price of a Toyota Corolla with 2018 year model, 100 000 km driven, and a manual transmission. Include 95% prediction limits.